

**A WEB-BASED RECRUITMENT MANAGEMENT
SYSTEM FOR THE NATIONAL COMPETITIVENESS
COMMISSION (NCC)**



**Isaac Bvunzawabaya (MIA281)
Charles Bere (MIA253)
BIS 2.2**

A WEB-BASED RECRUITMENT MANAGEMENT SYSTEM FOR THE NATIONAL COMPETITIVENESS COMMISSION

(NCC)

Commented [SM1]: Use lower case as the rest of this page

By

Isaac Bvunzawabaya and Charles Bere



Submitted in partial fulfilment of the requirements for the

BCom. Honours Degree in Information Systems

Department of Information and Marketing Sciences in the

Faculty of Business Sciences at the

Midlands State University

Gweru

Zimbabwe

November 2026

Supervisor: S.Mataga

CHAPTER ONE: INTRODUCTION

1.1 Introduction

This project focuses on the development of a Web Based Online Recruitment Management System for the National Competitiveness Commission (NCC). The proposed system seeks to automate and streamline the recruitment process by providing a centralized online platform for advertising vacancies, receiving applications, managing candidate information, conducting shortlisting, and communicating with applicants.

Currently, the Commission relies heavily on manual recruitment procedures, which involve receiving applications through email and physical submissions. As the volume of applications increases, the process becomes time-consuming and difficult to manage efficiently. The proposed system aims to improve the effectiveness of recruitment activities by reducing paperwork, improving candidate tracking, enhancing communication, and providing timely recruitment reports. This chapter presents the background of the study, the organisational context, the problem statement, project aim, objectives, methods and instruments, project justification, and conclusion.

1.2 Background of the Study

Human resources are among the most valuable assets in any organisation. The ability to recruit competent and qualified personnel plays a significant role in determining the success of an organisation. As technology continues to transform business operations, organisations are increasingly adopting digital solutions to improve efficiency and service delivery.

Recruitment is one area that has greatly benefited from technological advancements. Web Based Online recruitment systems have enabled organisations to simplify application management, improve communication with candidates, reduce administrative costs, and enhance transparency throughout the hiring process.

At the National Competitiveness Commission, recruitment activities are currently conducted through a largely manual process. When vacancies arise, job advertisements are

published through social media platforms and other communication channels. Interested candidates submit their applications either through email or by physically delivering application documents to the Commission's offices.

Applications received through email are first downloaded and printed, while hand-delivered applications are removed from their envelopes and prepared for processing. The printed electronic applications and physically submitted applications are then combined into one pool. Human Resources and Administration personnel subsequently conduct the shortlisting process manually by reviewing each application against the vacancy requirements.

After the shortlisting exercise has been completed, shortlisted candidates are contacted individually through telephone calls and text messages to notify them about interviews and other recruitment-related matters.

While this process has been functional over the years, it presents several challenges. The printing and sorting of applications consume significant amounts of time and resources. Tracking applications can be difficult, particularly when dealing with large numbers of applicants. The manual shortlisting process increases the risk of delays and inconsistencies, while communicating with applicants individually requires considerable effort from Human Resources personnel. Furthermore, generating recruitment statistics and reports becomes a labour-intensive task.

As the National Competitiveness Commission continues to grow and modernise its operations, there is a need to introduce a system that improves the management of recruitment activities. A Web Based Online Recruitment Management System would provide a centralized platform where vacancies can be advertised, applications submitted electronically, candidate information managed, and communication conducted efficiently. This would reduce administrative workload and improve the overall effectiveness of the recruitment process.

1.2.1 Background of the Organisation

The National Competitiveness Commission (NCC) is a statutory body established by an Act of Parliament — the National Competitiveness Commission Act, Chapter 14:36 (No. 6/2017), which came into effect on 23 June 2017. The Commission falls under the purview of Zimbabwe's Ministry of Industry and Commerce and is headquartered in Marlborough, Harare. The mandate of the Commission, as guided by the Act, is to enable the creation of a competitive environment for Zimbabwean business through the development, coordination, and implementation of key policy improvements.

The Commission works closely with Government Ministries, the private sector, academia, development partners, and other stakeholders to identify barriers affecting competitiveness and recommend measures aimed at improving productivity, efficiency, and economic growth.

As part of fulfilling its mandate, NCC requires competent personnel with the necessary skills and expertise. Consequently, effective recruitment processes are essential to ensure that the organisation attracts and selects the best candidates for available positions.

1.2.2 Organisational Structure

The NCC is governed by a Board of Commissioners drawn from various sectors of the economy, providing strategic direction and oversight. The Board reports to the Minister responsible for Industry and Commerce, ensuring that the Commission's activities remain aligned with national development goals. Operationally, the Commission is organised into core departments, including the Competitiveness Department, which provides evidencebased analysis of supply and demand across sectors of the economy and analysing the country's global competitiveness performance through engagement with international competitiveness institutions; the Research and Policy Analysis Department, which provides evidence-based research and regulatory analysis to inform productivity and competitiveness policy. The other departments are the Business Development, Communications, Finance, Procurement, Audit and Compliance.

process become especially consequential: any delay or error in identifying and onboarding the right talent has a direct bearing on the Commission's ability to deliver on its mandate.

Currently, the NCC manages recruitment through manual processes, including paper forms, email applications, and spreadsheets.

This approach has led to lost applications, delayed communication with candidates, and difficulty in tracking applicant status throughout the hiring process. The lack of a centralised system increases the administrative workload on HR staff and reduces the overall efficiency of selecting qualified candidates, a challenge that directly motivates the problem addressed in this study.

1.3 Problem Statement

The National Competitiveness Commission currently relies on a manual recruitment process that results in inefficient application management, time-consuming candidate processing, and delays in recruitment activities.

1.4 Project Aim

The aim of this project is to develop a Web Based Recruitment Management System that centralises and automates the recruitment process at the National Competitiveness Commission, from job posting to candidate selection.

1.5 Objectives

The system should be able to:

1. Avail a centralised platform that allows HR and Admin users to post job vacancies and applicants to view and apply for them online, ~~eliminating reliance on paper forms and email-based applications within the first release of the system.~~
2. Automatically track and update the status of every application as it moves through the recruitment pipeline (~~Applied, Shortlisted, Interview, Hired,~~

Commented [SM2]: Your objectives are too long. Remove the deleted/highlighted areas

~~Rejected), enabling HR staff to retrieve the current status of any applicant within seconds rather than searching through spreadsheets.~~

3. Automatically score and rank applicants against job requirements using a text-similarity algorithm (TF-IDF and cosine similarity) applied to submitted CVs, ~~so that HR staff can identify the top-matching candidates for a vacancy without manually reading every CV.~~
4. Notify applicants automatically by email whenever their application status changes, reducing delays in candidate communication that currently result from manual follow-up.
5. Maintain a complete, time-stamped audit log of every status change made on an application, ~~ensuring accountability and traceability of hiring decisions made by HR and Admin users.~~

1.6 Methods and Instruments

1.6.1 Software Tools

The system will be developed using React for the frontend, Django and Django REST Framework for the backend, and MySQL as the database management system. Authentication will be implemented using JSON Web Tokens (JWT). The CV-matching functionality will be implemented using Python's scikit-learn library for TF-IDF vectorisation and cosine similarity computation, pdfplumber for extracting text from uploaded PDF CVs, and rapidfuzz for fuzzy duplicate-application detection.

1.6.2 Data Gathering Techniques

Interviews

Interviews will be conducted with Human Resources and Administration personnel to understand the current recruitment process and gather system requirements.

Observation

Direct observation of recruitment activities will be carried out to identify weaknesses and inefficiencies in the existing process.

Document Analysis

Recruitment-related documents such as vacancy notices, application forms, interview schedules, and recruitment reports will be analysed to understand information requirements.

Questionnaires

Questionnaires may be distributed to users and stakeholders to gather additional information regarding system expectations and requirements.

1.6.3 Methodology Used

The project will adopt the Waterfall Software Development Methodology. The methodology consists of the following stages:

- ✦ Requirements Analysis
- ✦ System Design
- ✦ System Development
- ✦ System Testing
- ✦ System Implementation
- ✦ System Maintenance

The Waterfall approach is suitable because it provides a structured framework that allows requirements to be gathered and documented before development begins.

1.7 Project Justification

The development of a Web based Online Recruitment Management System is justified by the need to improve efficiency, accuracy, transparency, and accountability in recruitment activities at the National Competitiveness Commission.

The proposed system will eliminate the need for printing and manually sorting applications, thereby reducing administrative costs and paperwork. It will improve candidate tracking, enhance communication between applicants and the Commission, and reduce the time required to conduct recruitment exercises.

The system will also provide a centralized repository for recruitment information, making it easier to generate reports and monitor recruitment performance. Ultimately, the proposed solution will contribute to improved service delivery and support the Commission in attracting and selecting qualified personnel more effectively.

1.8 Conclusion

This chapter introduced the proposed Web Based Online Recruitment Management System for the National Competitiveness Commission. It discussed the background of the study, provided information about the organisation, and explained the challenges associated with the current recruitment process. The chapter also presented the problem statement, project aim, SMART objectives, methods and instruments to be used during the study, and the justification for developing the proposed system. The next chapter reviews related literature and existing recruitment systems relevant to the study.

CHAPTER TWO: PLANNING PHASE

2.1 Introduction

The planning phase is one of the most important stages in the System Development Life Cycle (SDLC). It provides an opportunity for the organisation and the development team to assess whether the proposed system is practical, beneficial and capable of meeting the identified business requirements before resources are committed to development.

Following the challenges identified in Chapter One regarding the manual recruitment process at the National Competitiveness Commission (NCC), this chapter evaluates the viability of developing the proposed Web Based Online Recruitment Management System. It discusses the business value that the system will create for both internal and external stakeholders, analyses the feasibility of implementing the system, identifies potential risks that may affect project success together with mitigation strategies, and presents the proposed work plan based on the Waterfall Software Development Methodology.

2.2 Business Value

Business value refers to the measurable benefits that an organisation and its stakeholders derive from implementing a new information system. The proposed Web Based Online Recruitment Management System is expected to improve recruitment efficiency while creating value for government, management, employees, job applicants and other stakeholders associated with the National Competitiveness Commission.

2.2.1 Shareholder Value

The National Competitiveness Commission is accountable to several government institutions, including the **Ministry of Industry and Commerce**, the **Office of the President and Cabinet (OPC)**, **Parliament**, the **NCC Board**, and the **Ministry of Finance and Economic Development**.

The proposed system will provide significant value to these stakeholders by improving accountability, transparency and governance during recruitment exercises. Since all recruitment records will be stored electronically, management will be able to generate

accurate recruitment reports, monitor recruitment progress and demonstrate compliance with government procedures.

The system will also reduce operational costs associated with printing hundreds of applications, physical storage of documents and repetitive administrative work. These savings enable the Commission to utilise financial resources more effectively.

2.2.2 Customer Value

The primary customers of the proposed system are job applicants seeking employment opportunities at the National Competitiveness Commission.

Currently, applicants submit applications through email or by physically delivering documents to the Commission. After submission, applicants often have limited visibility regarding the progress of their applications.

The proposed Web Based Online Recruitment Management System will significantly improve the applicant's experience by allowing candidates to:

- ✦ Create secure online profiles.
- ✦ Submit applications electronically.
- ✦ Upload supporting documents online.
- ✦ Receive automatic confirmation once applications have been submitted.
- ✦ Receive interview invitations and recruitment updates electronically.
- ✦ Track the status of their applications without making enquiries to Human Resources.

These improvements will make the recruitment process more transparent, efficient and convenient for applicants.

2.2.3 Managerial Value

Management relies on timely and accurate information when making recruitment decisions.

The proposed system will assist the Executive Director, Human Resources and Administration Department, Departmental Managers and the NCC Board by providing:

- ✦ Automated shortlisting support.
- ✦ Recruitment statistics and dashboards.
- ✦ Vacancy management reports.
- ✦ Candidate tracking reports.
- ✦ Faster access to applicant information.
- ✦ Improved decision-making through real-time information.

Instead of manually sorting applications and compiling reports, managers will have immediate access to recruitment information whenever required.

2.2.4 Employee Knowledge

The implementation of the Web Based Online Recruitment Management System will improve the digital skills of employees involved in recruitment activities.

Human Resources personnel will gain practical experience in:

- ✦ Electronic recruitment management.
- ✦ Digital document management.
- ✦ Online communication with applicants.
- ✦ Report generation.
- ✦ Data security and confidentiality.
- ✦ Information management using modern technologies.

This contributes to continuous professional development while reducing dependence on manual administrative procedures.

2.2.5 Supplier Value

Several suppliers provide services that support the Commission's ICT infrastructure.

These include:

- ✦ Website hosting provider (Liquid).
- ✦ Domain registration service provider (Liquid).
- ✦ SMS notification providers.
- ✦ Internet service provider (Liquid).
- ✦ ICT equipment suppliers.
- ✦ VoIP Service Provider (Liquid)

Implementation of the proposed system will strengthen collaboration between NCC and these suppliers through continued utilisation of hosting, maintenance, internet connectivity and communication services required to support the online recruitment platform.

2.2.6 Channel Partner Value

The National Competitiveness Commission works closely with numerous strategic partners that contribute towards national competitiveness.

These include:

- ✦ Ministry of Higher and Tertiary Education
- ✦ Ministry of Foreign Affairs and International Trade
- ✦ Business Regulators and Support Organisations (CTC, CZI, CPC, CCZ, ZIDA, RBZ, ZNCC, BAZ, ZimTrade, ZTA, ZIMRA, SECZ and the Chamber of Mines)
- ✦ Academia
- ✦ Media
- ✦ Development Partners

- ✦ Civil Society Organisations
- ✦ Standards Development Fund

The proposed system will ensure that the Commission recruits highly qualified personnel more efficiently, thereby strengthening service delivery and enhancing collaboration with these strategic partners. Efficient recruitment also enables the Commission to implement programs and projects within expected timeframes.

2.2.7 Societal Value

The proposed Web Based Online Recruitment Management System will provide several benefits to society.

These include:

- ✦ Promoting fair and transparent recruitment practices.
- ✦ Providing equal employment opportunities to applicants regardless of geographical location.
- ✦ Reducing paper consumption, thereby supporting environmental sustainability.
- ✦ Improving public confidence in recruitment processes.
- ✦ Supporting Zimbabwe's national digital transformation agenda.
- ✦ Improving service delivery through efficient recruitment of qualified personnel.

Ultimately, society benefits when public institutions recruit competent employees in an efficient, transparent and accountable manner.

2.2.8 Stakeholder Analysis

One important activity during the planning phase is identifying stakeholders and understanding how they will be affected by the proposed system. The Web based Online

Recruitment Management System will benefit both internal and external stakeholders by improving communication, transparency, and the efficiency of recruitment processes.

Table 2.1 Stakeholder Analysis

Stakeholder	Interest in the Project	Influence	Expected Benefit
Ministry of Industry and Commerce	High	High	Improved governance and accountability through accurate recruitment reporting.
NCC Board	High	High	Timely recruitment reports and improved oversight of recruitment activities.
Office of the President and Cabinet (OPC)	Medium	High	Improved compliance with Government recruitment procedures.
Ministry of Finance and Economic Development	Medium	High	Better utilisation of financial resources through reduced recruitment costs.
Ministry of Higher and Tertiary Education	Medium	Medium	Enhanced collaboration through recruitment of competent professionals.

Commented [SM3]: Try to reduce spaces in your tables as done below and also separate your tables so that they do not overlap to the next page

Stakeholder	Interest in the Project	Influence	Expected Benefit
Ministry of Foreign Affairs and International Trade	Low	Medium	Improved institutional capacity through effective staffing.
Business Regulators and Support Organisations	Medium	Medium	Better service delivery resulting from efficient recruitment.

Academia	Medium	Low	Easier access to employment opportunities for graduates and researchers.
Media	Medium	Medium	Improved dissemination of employment opportunities and organisational information.
Parliament	Low	High	Increased accountability and transparency in recruitment processes.
Development Partners	Medium	Medium	Strengthened institutional capacity to implement funded programs.
Civil Society Organisations	Low	Low	Greater public confidence in fair recruitment practices.
Standards Development Fund	Medium	Medium	Improved project implementation through timely recruitment of skilled personnel.
Stakeholder	Interest in the Project	Influence	Expected Benefit
Staff	High	High	Reduced workload and improved efficiency in recruitment administration.
Job Applicants	Very High	Low	Faster, fairer and more transparent recruitment services.

2.2.9 SWOT Analysis

Before developing a new information system, it is important to assess the organisation's internal and external environment.

Table 2.2 SWOT Analysis

Strengths	Weaknesses
Existing ICT Department	Manual recruitment process
Reliable Internet Connectivity	Heavy dependence on paperwork
Management support	Time-consuming shortlisting
Existing Website	Manual communication with applicants https://www.ncc-zim.co.zw/

The SWOT analysis demonstrates that the strengths and opportunities available to the Commission outweigh the identified weaknesses and threats, making the implementation of the proposed system viable.

Enhanced Economic Feasibility

Table 2.3 Estimated Project Costs

Item	Year 1 (USD)	Year 2 (USD)	Year 3 (USD)	Total (USD)
Software Development (one-time)	2,500	–	–	2,500
Domain & Hosting	150	150	150	450
SSL Certificate	80	80	80	240
Staff Training (one-time)	300	–	–	300
System Testing (one-time)	250	–	–	250
Documentation (one-time)	150	–	–	150

Maintenance & Support	300	300	300	900
Contingency (one-time)	400	–	–	400
Total Estimated Cost	4,130	530	530	5,190

Table 2.4 Expected Benefits

Tangible Benefits	Intangible Benefits
Reduced printing costs	Improved organisational image
Reduced stationery costs	Increased applicant satisfaction
Reduced storage costs	Better decision-making
Reduced communication costs	Improved transparency
Reduced recruitment time	Enhanced service delivery

Table 2.5 Three (3) Year Estimated Benefits

Tangible Benefit	Year 1 (USD)	Year 2 (USD)	Year 3 (USD)	Total (USD)
Reduced printing costs	300	330	360	990
Reduced stationery costs	200	220	240	660
Reduced storage costs	150	170	190	510
Reduced communication costs	280	310	340	930
Reduced recruitmentrelated labour costs	700	800	900	2,400
Total Estimated Benefit	1,630	1,830	2,030	5,490

Table 2.6 Cost-Benefit Analysis

Current Manual System	Proposed Online System
Printing every CV received	Electronic document management
Physical filing cabinets	Digital database
Manual shortlisting	Automated screening support
Telephone calls	Automated email and SMS notifications
Manual report preparation	Instant report generation

The analysis indicates that although the proposed system requires an initial investment, the long-term operational savings and efficiency gains justify the cost.

Table 2.7 3-Year Cost-Benefit Summary

Year	Estimated Cost (USD)	Estimated Benefit (USD)	Net Benefit (USD)	Cumulative Net Benefit (USD)
Year 1	4,130	1,630	(2,500)	(2,500)
Year 2	530	1,830	1,300	(1,200)
Year 3	530	2,030	1,500	300
3-Year Total	5,190	5,490	300	—

As shown above, the proposed system requires a heavier initial outlay in Year 1, covering one-time development and setup costs. From Year 2 onward, only recurring hosting, SSL and maintenance costs are incurred, amounting to approximately USD 530 per year, while estimated tangible savings continue to grow as adoption increases. On a cumulative basis, the direct cost savings alone are projected to recover the total investment early in Year 3, meaning the system is financially self-sustaining within the project's 3-year horizon. This margin is modest when measured in dollar terms alone, but the tangible savings do not capture the full value of the system. As a public institution accountable to Parliament, the Office of the President and Cabinet, and the Ministry of Finance and Economic Development, the Commission stands to gain substantially from the intangible benefits identified earlier in this chapter, including improved transparency, stronger governance, better decision-making and enhanced applicant satisfaction. Taken together with the low

recurring cost of USD 530 per year beyond Year 1, these factors support the conclusion that the proposed system is both economically and strategically justified.

2.3 Feasibility Analysis

Feasibility analysis determines whether the proposed system can be successfully developed and implemented.

2.3.1 Technical Feasibility

The proposed system is technically feasible.

The National Competitiveness Commission already possesses much of the ICT infrastructure required to support the proposed solution. The organisation has internet connectivity, Local Area Networks (LANs), computer workstations and an Information and Communication Technology Department capable of providing technical support.

The proposed system will be developed using modern web technologies and hosted on the Commission's existing web infrastructure. Users will access the system through standard web browsers without requiring specialised software.

2.3.2 Economic Feasibility

Although financial resources will be required during development, implementation and maintenance, the long-term benefits outweigh the associated costs.

Table 2.8 Economic Feasibility

Estimated Costs	Expected Benefits
Software development	Reduced printing costs
Staff training	Reduced stationery usage
System hosting	Faster recruitment process

Maintenance	Reduced administrative workload
Internet services	Improved reporting
Technical support	Better record management

The reduction in paper usage, printing costs and manual labour will generate long-term cost savings for the Commission.

2.3.3 Social Feasibility

The proposed system is socially acceptable.

Employees are already familiar with using computers and email during recruitment activities. Applicants increasingly expect organisations to provide online application platforms.

Appropriate training and user support will ensure a smooth transition from the existing manual recruitment process to the proposed online system.

2.3.4 Operational Feasibility

Operational feasibility assesses whether the proposed system will function effectively within the organisation.

The proposed Online Recruitment Management System aligns with the Commission's existing recruitment procedures while simplifying several manual activities.

Human Resources personnel will continue to manage recruitment activities, but many repetitive tasks such as receiving applications, sorting documents, communicating with applicants and generating reports will become automated.

Management support, existing ICT infrastructure and user training indicate that successful implementation is operationally achievable.

2.4 Risk Analysis

The successful implementation of the proposed system depends on effective management of potential risks.

Table 2.9 Risk Analysis

Risk	Likelihood	Impact	Mitigation Strategy
Internet connectivity failure	Medium	High	Provide backup internet connectivity.
Resistance to change	Medium	Medium	Conduct user training and awareness programmes.
Data loss	Low	High	Implement daily database backups.
Cybersecurity attacks	Medium	High	Install firewalls, SSL encryption and strong authentication.
Risk	Likelihood	Impact	Mitigation Strategy
Power outages	Medium	Medium	Install UPS devices and backup power solutions.
System downtime	Low	High	Perform scheduled maintenance and continuous monitoring.
Human error	Medium	Medium	Provide user manuals and ongoing technical support.

Table 2.10 Risk Heat Map

Impact / Probability	Low	Medium	High
High Impact	Data Loss	Internet Failure	Cyber Attacks
Medium Impact	Human Error	Power Failure	Resistance to Change
Low Impact	Hardware Failure	Staff Turnover	Minor Software Bugs

2.5 Work Plan

The project will follow the Waterfall Software Development Methodology.

Table 2.11 Proposed Activity Schedule

Activity	Jul	Aug	Sep	Oct	Nov
Requirements Gathering	✓				
System Analysis	✓	✓			
System Design		✓			
System Development		✓	✓		
System Testing			✓		
System Implementation				✓	
User Training				✓	
Documentation	✓	✓	✓	✓	✓

Commented [SM4]: On this table you can give us the actual start date and stop date and duration as columns rather than months

Project Evaluation				✓
--------------------	--	--	--	---

Table 2.12 Gantt Chart

Activities	Jul	Aug	Sep	Oct	Nov
Requirements Gathering	■				
Feasibility Study	■				
Analysis	■	■			
Design		■			
Coding		■	■		
Testing			■		
Implementation				■	
User Training				■	
Documentation	■	■	■	■	■
Maintenance					■

2.5.1 Project Milestones

Table 2.13 Project Milestones

Milestone	Expected Completion
-----------	---------------------

Requirements Approved	July 2026
Analysis Completed	August 2026
System Design Approved	August 2026
Development Completed	September 2026
Testing Completed	October 2026
System Go-Live	October 2026
Final Documentation Submitted	November 2026

2.5.2 Deliverables

The following deliverables are expected upon completion of the project:

- ✦ Approved System Proposal
- ✦ Software Requirements Specification (SRS)
- ✦ System Design Document
- ✦ Database Design
- ✦ Online Recruitment Management System
- ✦ User Manual
- ✦ Administrator Manual
- ✦ Test Report
- ✦ Final Project Documentation

2.6 Conclusion

This chapter examined the planning phase of the proposed Web Based Online Recruitment Management System. It demonstrated the business value that the system will provide to the

National Competitiveness Commission and its stakeholders, evaluated the technical, economic, social and operational feasibility of the project, identified potential implementation risks together with mitigation strategies, and presented the project work plan based on the Waterfall Software Development Methodology. The next chapter will focus on the analysis phase, where the functional and non-functional system requirements, process models and system specifications will be developed.

CHAPTER THREE: ANALYSIS PHASE

3.1 Introduction

This chapter presents the analysis phase of the proposed Web Based Recruitment Management System for the National Competitiveness Commission (NCC). It documents the data gathering techniques employed during the study, examines the existing manual recruitment process, identifies its weaknesses, evaluates the alternatives available for addressing the identified problem, and presents the functional and non-functional requirements of the proposed system.

3.2 Data Gathering Techniques

To gain an accurate understanding of the current recruitment process and the requirements of the proposed system, four data gathering techniques were employed: interviews, observation, document analysis, and questionnaires.

Interviews

Interviews were conducted with Human Resources and Administration personnel at the NCC to understand how vacancies are advertised, how applications are received and processed, and how shortlisting decisions are currently made. The interviews provided first-hand insight into the day-to-day challenges faced by staff in managing the recruitment process manually, and helped clarify the recruitment pipeline stages that the proposed system needed to support.

Observation

Direct observation of recruitment activities was carried out over a period of time, allowing the researchers to witness first-hand information on how applications are received, sorted, and shortlisted, and how communication with applicants is currently handled. Observation was useful for identifying inefficiencies and bottlenecks that were not always explicitly raised during interviews, such as the time spent printing and physically sorting applications.

Commented [SM5]: Include advantages and disadvantages of each technique used and possible outcomes/findings on each technique

Document Analysis

Recruitment-related documents, including vacancy notices, application forms, interview schedules, and previously compiled recruitment reports, were reviewed to understand the type and structure of information currently captured and processed. This analysis helped establish the inputs, processes, and outputs of the existing system described in section 3.3.

Questionnaires

Questionnaires were distributed to Human Resources staff and a sample of recent applicants to gather additional information regarding the challenges experienced with the current process and expectations for a new system. Responses reinforced the concerns raised during interviews, particularly regarding delays in communication and difficulty tracking application status.

3.3 Analysis of the Existing System

3.3.1 General Description of the Existing System

The NCC currently manages recruitment through a largely manual process. When a vacancy arises, job advertisements are published through social media platforms and other communication channels. Interested candidates submit their applications either by email or by physically delivering application documents to the Commission's offices. Applications received by email are downloaded and printed, while hand-delivered applications are removed from their envelopes and prepared for processing. The printed electronic applications and the physically submitted applications are then combined into a single pool, from which Human Resources and Administration personnel manually review each application against the vacancy requirements. Shortlisted candidates are then contacted individually by telephone or text message to notify them of interviews and other recruitment-related matters, and recruitment statistics are subsequently compiled by hand.

3.3.2 Inputs of the Existing System

- ✦ Vacancy details (job title, requirements, closing date) submitted by department heads
- ✦ Applications comprising a CV, cover letter, certified copies of academic certificates, and a national identification document, submitted by email or physical delivery
- ✦ Applicant's contact details (phone number and email address)

3.3.3 Processes of the Existing System

- ✦ Advertising vacancies on social media and other communication channels
- ✦ Receiving applications by email and physical delivery
- ✦ Printing of email-submitted applications
- ✦ Combining printed and physically submitted applications into a single pool
- ✦ Manually reviewing and shortlisting applications against vacancy requirements
- ✦ Contacting shortlisted applicants by telephone or SMS
- ✦ Conducting interviews
- ✦ Manually compiling recruitment reports and statistics

3.3.4 Output of the Existing System

- ✦ A shortlist of candidates for each vacancy
- ✦ Interview schedules
- ✦ Verbal or SMS interview notifications sent to applicants
- ✦ Recruitment reports compiled manually, typically in spreadsheet or word-processed form

3.3.5 Activity Diagram of the Existing System

Figure 3.1 below illustrates the sequence of activities involved in the current manual recruitment process, from the advertising of a vacancy through to the compilation of a recruitment report.

Commented [SM6]: Your diagram can fit in here on this page as shown here

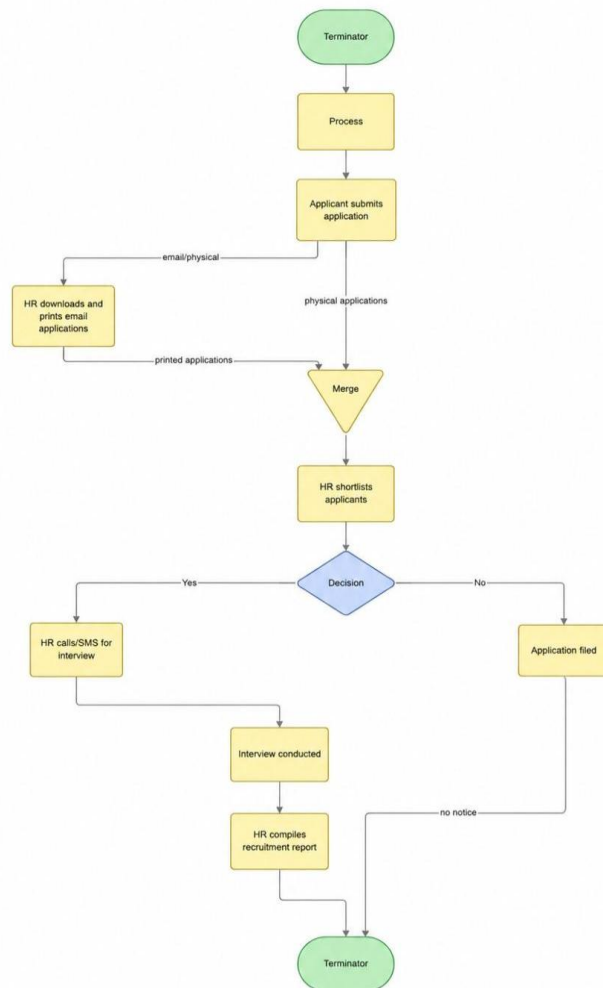


Figure 3.1: Activity Diagram of the Existing Manual Recruitment Process

3.3.6 Context Diagram of the Existing System

Figure 3.2 presents the context diagram (Level 0 Data Flow Diagram) of the existing system, showing the manual recruitment process as a single process interacting with its external entities: the Applicant, HR & Admin Personnel, and Management.

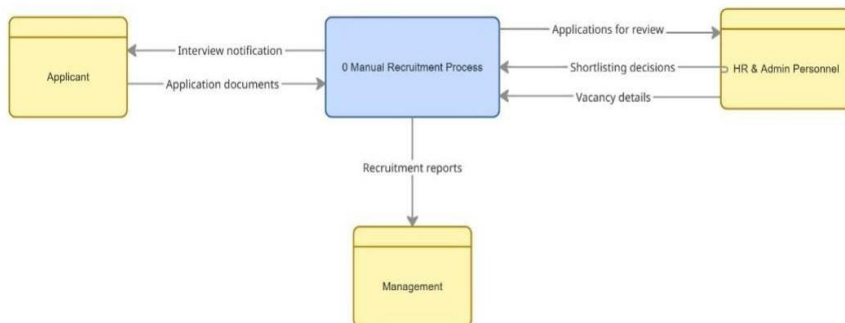


Figure 3.2: Context Diagram (Level 0 DFD) of the Existing System

3.3.7 Data Flow Diagram of the Existing System

Figure 3.3 decomposes the context diagram into a Level 1 Data Flow Diagram, showing the individual processes, data stores, and data flows that make up the current manual recruitment process.

Commented [SM7]: Enlarge the diagram

Commented [SM8]: Move this to next page where your diagram is.

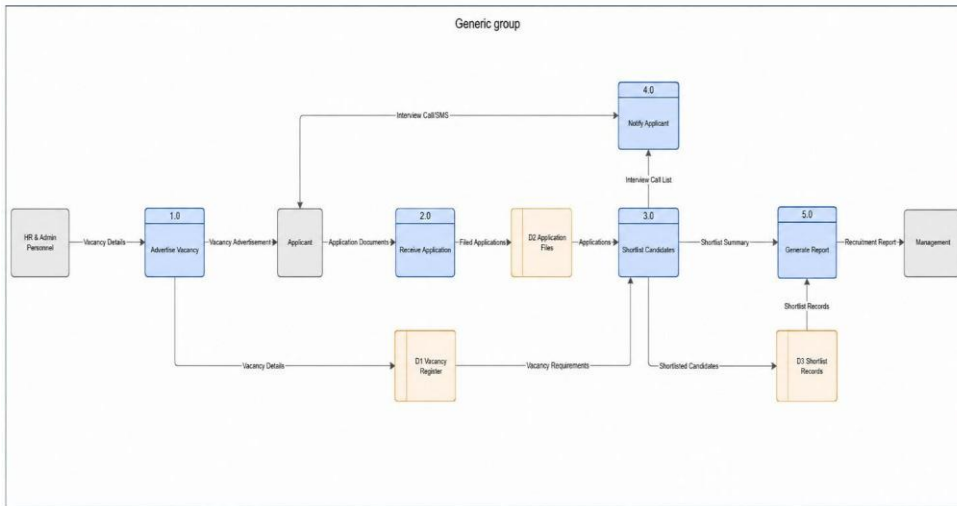


Figure 3.3: Level 1 Data Flow Diagram of the Existing System

Commented [SM9]: Enlarge your diagram and increase font on text. The text/labelling is not clear am failing to read

3.4 Weaknesses of the Existing System

The analysis of the existing system, informed by the interviews, observation, document analysis and questionnaires described in section 3.2, revealed the following weaknesses:

- ✦ Time-consuming: significant staff time is spent printing, sorting, and combining applications before shortlisting can even begin
- ✦ Risk of lost or misplaced applications, particularly physical copies handled manually
- ✦ Inconsistent shortlisting: there is no standardised scoring mechanism, making manual review prone to human error and inconsistency
- ✦ Delayed and labour-intensive communication: notifying applicants individually by telephone or SMS is time-consuming for HR staff and communication can be inconsistent

- ✦ Difficulty tracking applicant status: there is no centralised way for HR staff or applicants to check where an application stands in the recruitment pipeline
- ✦ Labour-intensive reporting: compiling recruitment statistics manually is slow and prone to error
- ✦ Lack of an audit trail: there is no systematic record of who reviewed or changed an application's status and when
- ✦ Poor scalability: the manual process does not scale well as the volume of applications increases

3.5 Evaluation of Alternatives

Three alternative approaches to addressing the weaknesses identified in section 3.4 were considered: outsourcing recruitment functions to a third party, improving the current manual system, and developing a new bespoke system.

3.5.1 Outsourcing

Outsourcing would involve engaging a third-party recruitment agency or an existing commercial recruitment platform to manage some or all of the Commission's recruitment activities. This option would reduce the internal development effort required and provide access to established, professionally supported tools. However, it introduces recurring subscription or service costs, raises data privacy concerns given that the Commission handles sensitive applicant and personnel information as a statutory body accountable to Parliament, and offers limited customisation to the Commission's specific recruitment pipeline and reporting needs. For these reasons, outsourcing was not selected.

3.5.2 Improvement of the Current System

This option involves making incremental improvements to the existing manual process, such as introducing shared spreadsheets or standard email templates, without developing a dedicated system. While low in cost and quick to implement, this approach does not resolve the fundamental issues identified in section 3.4, including the lack of centralised application tracking, the absence of an audit trail, and the lack of automated candidate

scoring. It would remain heavily reliant on manual effort and would not scale effectively as application volumes grow. For these reasons, improving the current system was considered a short-term stopgap rather than a solution to the underlying problem, and was not selected.

3.5.3 Development

The third alternative is the development of a bespoke Web Based Recruitment Management System tailored to the Commission's specific recruitment workflow, as outlined in Chapter One. This option offers full customisation to the Commission's recruitment pipeline (Applied, Shortlisted, Interview, Hired, Rejected) and reporting requirements, keeps sensitive applicant data under the Commission's direct control, and, as shown in the costbenefit analysis in Chapter Two, is more cost-effective over the medium term than recurring outsourcing fees. Development was therefore selected as the most viable alternative and forms the basis of the requirements analysis presented in section 3.6.

3.6 Requirements Analysis of the New System

3.6.1 Functional Requirements

Based on the objectives outlined in Chapter One and the weaknesses identified in section 3.4, the following functional requirements were derived for the proposed system:

Table 3.1: Functional Requirements

ID	Functional Requirement
FR1	The system shall allow HR and Admin users to register, log in, and post job vacancies online.
FR2	The system shall allow applicants to register, log in, and apply for vacancies electronically by submitting a CV and supporting documents.
FR3	The system shall automatically track and update the status of every application as it moves through the recruitment pipeline (Applied, Shortlisted, Interview, Hired, Rejected).

ID	Functional Requirement
FR4	The system shall automatically score and rank applicants against job requirements using a TF-IDF and cosine-similarity algorithm applied to submitted CVs.
FR5	The system shall automatically notify applicants by email whenever their application status changes.
FR6	The system shall maintain a complete, time-stamped audit log of every status change made to an application.
FR7	The system shall allow HR and Admin users to generate recruitment reports.

Figure 3.4 presents a use case diagram illustrating the interactions between the Applicant, HR Officer, and Admin actors and the main functions of the proposed system.

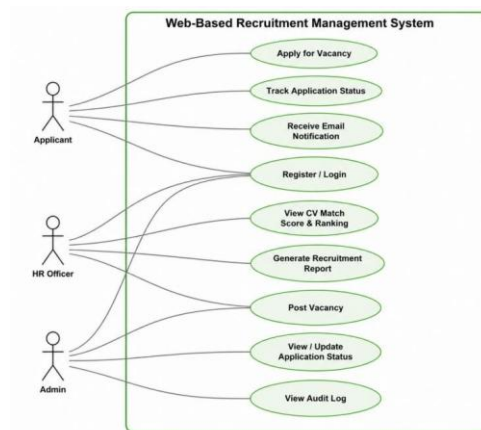


Figure 3.4: Use Case Diagram of the Proposed System

3.6.2 Non-functional Requirements

- ⚙ Performance: the system should respond to routine user actions, such as searching or filtering applications, within an acceptable time (e.g. under 3 seconds under normal load).

- ✦ Security: user passwords shall be securely hashed, authentication shall be implemented using JSON Web Tokens (JWT), and role-based access control shall restrict functionality according to the Applicant, HR, and Admin roles.
- ✦ Usability: the interface shall be intuitive for HR staff and applicants with varying levels of digital literacy.
- ✦ Availability: the system shall be accessible online at all times to allow applicants to apply for vacancies at their convenience.
- ✦ Scalability: the system shall be able to handle increasing volumes of applications and concurrent users without significant degradation in performance.
- ✦ Maintainability: the system shall be built using modular, well-documented code to facilitate future maintenance and enhancement.
- ✦ Data Integrity: input validation shall be enforced throughout the system, and fuzzy duplicate detection shall be used to identify duplicate applications.

3.7 Conclusion

This chapter presented the analysis phase of the proposed Web Based Recruitment Management System. It outlined the data gathering techniques used during the study, analysed the existing manual recruitment process including its inputs, processes, outputs and associated weaknesses, and evaluated outsourcing, improvement of the current system, and development as alternative solutions, justifying the selection of in-house development. The chapter concluded with a presentation of the functional and non-functional requirements of the proposed system. The next chapter focuses on the design phase, covering the system's architecture, database design, and user interface design.